Fiscal Year:	FY 2005	**Task Last Updated:**	FY 07/10/2006
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Project Title:	Cell & Molecular Biomechanics: Cardiac & Skeletal Muscle		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Muscle Alterations and Atrophy Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
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No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Regnier, Michael (University of Washington)		
Grant/Contract No.:	MA.002.11		
Performance Goal No.:			
Performance Goal Text:			
	The major goal of “Cell & Molecular Biomechanics: Cardiac & Skeletal Muscle” is to produce a muscle cell model that will: explain biomechanical adaptations that occur with alterations in muscle protein isoforms due to changes in activity level, as in prolonged space travel; predict bioenergetic changes associated with changes in activity level; be integrated into computational models of human limb and heart. It is a collaborative project between Florida State University and the University of Washington. 1. The original Aims of the project are: a. Identify contractile protein composition of skeletal and cardiac muscles from high-activity and low-activity rats. b. Characterize contractile properties (phenotype) of selected muscles containing unique mixtures of protein isoforms, as identified in Aim 1. c. Develop a “digital” cell biomechanical model. 2. Key Findings during the current funding period: a. Nano-mechanical model of the muscle sarcomere (a component of the Digital Astronaut): - Presence of filament compliance could modulate the dependence of steady state isometric force and the rate of tension development on thin filament activation level that is determined by calcium in the living cell. - Tuning of biomechanics through compliance at the protein level—-as demonstrated in our		

Task Description:	studies—-may be as significant as biomechanical tuning at higher levels of organization—-as demonstrated by Prof. Tom McMahon and colleagues' ability to improve human runners' performance through altering the characteristics of the track surface. The influence of compliance (or its inverse, elasticity) thus needs to be considered in biomechanical components at all levels of organization in physiological models to describe changes in muscle associated with exercise and microgravity. b. Signaling pathways in muscle plasticity: - The immunosuppressant drug rapamycin has little or no effect on maximum calcium-activated force, but (at some concentrations) increases calcium-sensitivity of isometric force and inhibits the kinetics of tension redevelopment; this indicates that rapamycin affects calcium-regulatory proteins (troponin/tropomyosin) more than actomyosin. - Attenuation of cardiac hypertrophy by rapamycin is not due to direct effects of rapamycin on myofilament contractility. 3. Impact: a. Our computational model of muscle is the cellular and molecular component for larger, complex multiscale models such as the anticipated "digital astronaut." The "digital astronaut" will be a physiological simulation of an astronaut that will be adjustable to individual astronauts and can be used to predict the response of astronauts to physical and health challenges during and after long-term space travel. It will also be used for "in silico" testing during countermeasure development. b. To develop effective countermeasures against muscle loss and adverse changes in muscle phenotype during long-term space travel, it is necessary to understand both muscle degradation pathways (Goldberg project) and signaling pathways associated with muscle growth and hypertrophy. Our experiments and modeling are allowing us to address the latter. 4. Proposed Research Plan: This project has been completed. The biomechanical model developed in aim 3 is now being applied to problems in cardiovascular physiology and disease.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our experimental data and modeling have implications for several areas of basic biology and biomedicine on earth. 1. Athletic and physical performance. Implications for the muscular component of astronaut performance are also relevant to cellular and molecular aspects of athletic and physical performance on earth. This applies not only to competitive athletes, but also to patients with muscle weakness due to muscle weakness, metabolic diseases, or heart failure for example. 2. Functional significance of protein isoforms in muscle function. Multiple varieties of related protein isoforms co-exist in cells because multiple, related genes are turned on simultaneously (due to the diploid nature of the genome and also to gene families for many proteins, and also due to alternative splicing of mRNA). It is not well-established how the presence of multiple protein isoforms of contractile proteins affects muscle function. Our experiments and modeling allow us to address this basic issue. It is also relevant to disease states where additional isoforms are present in the disease state, or to diseases that result from mutations in contractile proteins which are, in the broadest sense, yet another "isoform" (albeit detrimental). One example of such a family of diseases in striated muscle are the familial hypertrophic cardiomyopathies. 3. Functional significance of protein compliance in muscle function. This is another basic science aspect of our studies. It is not yet known whether proteins with different biomechanical properties exert their influence primarily through biomechanics. It is possible that some disease-related destabilization of in myofilament protein structure do affect the overall biomechanics of the sarcomere. 4. Signaling pathways involved in muscle hypertrophy. Our data and models will allow us to begin evaluation of proposed signaling pathways that modulate gene expression associated with striated muscle hypertrophy. Different pathways are implicated in skeletal and cardiac muscle hypertrophy, and between physiological (i.e., exercise-induced) versus pathological hypertrophy. Development of countermeasures to muscle wasting and muscle weakening will require a clear understanding of pathways and control mechanisms for both muscle degradation and hypertrophy.
Task Progress:	We have made significant progress in all aspects of the project. This progress has been documented in several recent manuscripts and abstracts, with additional manuscripts in preparation.
Bibliography Type:	Description: (Last Updated: 02/14/2006)
Articles in Peer-reviewed Journals	Chase PB, Macpherson JM, Daniel TL. "A spatially explicit nanomechanical model of the half-sarcomere: myofilament compliance affects Ca(2+)-activation. " Ann Biomed Eng. 2004 Nov;32(11):1559-68. PMID: 15636115 , Nov-2004
Articles in Peer-reviewed Journals	Schoffstall B, Kataoka A, Clark A, Chase PB. "Effects of rapamycin on cardiac and skeletal muscle contraction and crossbridge cycling. " J Pharmacol Exp Ther. 2005 Jan;312(1):12-8. Epub 2004 Aug 11. PMID: 15306636 , Jan-2005